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Experiment 3 – Exploratory Data Analysis and Visualization

Aim

To perform exploratory data analysis (EDA) and data visualization on the Student Result dataset using Python.

Theory

Dataset Overview:

This dataset contains 512 student records and nine attributes covering demographic details, class information, and academic results. It follows the Result_Fact design from Experiment 1 and includes deliberately introduced missing entries, duplicate rows, unusually high study-hour values, and inconsistent class labels. These examples allow EDA to reveal common data-quality issues before model building. Including these irregularities also shows how quality checks can uncover problems that may affect later statistical analysis and predictive modeling.

Significance of EDA for Student Result Analytics:

EDA is an important stage in data analysis and model development because it helps to:

- **Data Cleaning & Quality Check:** Identifies missing values, inconsistencies, and outliers (e.g., extreme study_hours_per_week values).
- **Understanding Data Distribution:** Visualizes distributions (e.g., skewness in study_hours_per_week) to guide preprocessing steps.
- **Identifying Key Features:** Determines which features (e.g., attendance_pct) influence marks_obtained. Helps eliminate redundant variables.
- **Detecting Relationships & Patterns:** Correlation analysis reveals dependencies (e.g., between attendance_pct and marks_obtained). Useful for feature selection in machine learning.
- **Improving Model Interpretability:** Helps choose appropriate preprocessing techniques (e.g., normalization for skewed data). Assists in selecting models based on feature relationships.

1. Descriptive Analysis – Statistical Measures of Central Tendency

Central tendency measures indicate the "center" of the data distribution.

Includes:

- Mean: Represents the average value (e.g., typical study_hours_per_week).
- Median: The middle value, robust to outliers (e.g., for marks_obtained).
- Mode: The most frequent value, useful for categorical data (e.g., class).

Relevance:

- The average study_hours_per_week highlights typical study behavior.
- The median marks_obtained avoids skew from outliers.
- The mode of class shows the most common class among students.

2. Descriptive Analysis – Statistical Measures of Dispersion

Dispersion measures the spread of data.

Includes:

- Standard Deviation: Shows deviation from the mean.
- Variance: Quantifies spread (squared standard deviation).
- Range: Difference between maximum and minimum values.
- Interquartile Range (IQR): Range of the middle 50%, useful for outlier detection.

Relevance:

- High variance in study_hours_per_week indicates diverse study habits, and hints at data-entry outliers.
- IQR of marks_obtained helps identify typical performance and flags outliers.
- Standard deviation of attendance_pct shows consistency in how regularly students attend class.

3. Correlation Between Attributes

Correlation measures the strength of relationships between variables. The Pearson correlation coefficient (-1 to +1) helps determine:

- If attendance_pct and marks_obtained are correlated, students who attend more might score higher.
- If age correlates with marks_obtained, it could indicate a maturity-related performance pattern.

Relevance:

- Identifies key predictors for academic performance.
- Detects multicollinearity to remove redundant features.
- Guides feature selection for modeling.

4. Different Visualization Techniques & Their Uses

Visualizations help interpret patterns, detect anomalies, and support inferences. Suitable methods include:

- Histogram: Shows distributions (e.g., study_hours_per_week).
- Boxplot: Highlights outliers and comparisons (e.g., marks_obtained by class).
- Heatmap: Visualizes correlations.
- Scatter Plot: Analyzes relationships (e.g., attendance_pct vs. marks_obtained).

Output

Step 1: Loading the Dataset and Initial Exploration

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")

df = pd.read_csv('student_result_eda_dataset.csv')

print("--- Dataset Information ---")
print(df.info())
print("\n--- First 5 Rows ---")
print(df.head())
print("\n--- Missing Values Count ---")
print(df.isnull().sum())
```

```
--- Dataset Information ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 512 entries, 0 to 511
Data columns (total 9 columns):
 #   Column                Non-Null Count  Dtype  
---  ------                --------------  -----  
 0   student_id            512 non-null    int64  
 1   age                   480 non-null    float64
 2   class                 512 non-null    object 
 3   section               512 non-null    object 
 4   study_hours_per_week  486 non-null    float64
 5   attendance_pct        482 non-null    float64
 6   subject               512 non-null    object 
 7   city                  485 non-null    object 
 8   marks_obtained        512 non-null    float64
dtypes: float64(4), int64(1), object(4)
memory usage: 36.1+ KB
None

--- First 5 Rows ---
   student_id   age class section  study_hours_per_week  attendance_pct  \
0        1001   NaN  10th       B                  17.1            95.1   
1        1002  17.0  12th       C                  17.2             NaN   
2        1003  15.0  12th       A                  19.2             NaN   
3        1004  17.0   9th       C                   4.8            93.4   
4        1005  17.0  10th       B                  13.2            68.8   

   subject       city  marks_obtained
0  English    Chennai            95.5
1  Science      Delhi            67.3
2  English       Pune            50.0
3  History  Bangalore            37.9
4  History      Delhi            62.0

--- Missing Values Count ---
student_id              0
age                    32
class                   0
section                 0
study_hours_per_week    26
attendance_pct          30
subject                 0
city                    27
marks_obtained          0
dtype: int64
```

Dataset loaded — 512 entries, 9 columns; 32 missing in age, 26 in study_hours_per_week, 30 in attendance_pct, 27 in city.

Step 2: Task 1 – Descriptive Analysis – Statistical Measures of Central Tendency

```python
num_cols = ['age', 'study_hours_per_week', 'attendance_pct', 'marks_obtained']
cat_cols = ['class', 'section', 'subject', 'city']

central_tendency = {
    'Mean': df[num_cols].mean(),
    'Median': df[num_cols].median(),
    'Mode': df[cat_cols].mode().iloc[0]
}

print("=== Central Tendency Measures ===")
for measure, values in central_tendency.items():
    print(f"\n{measure}:")
    print(values)
```

```
=== Central Tendency Measures ===

Mean:
age                    14.956250
study_hours_per_week   15.212140
attendance_pct         74.999170
marks_obtained         68.319141
dtype: float64

Median:
age                    15.00
study_hours_per_week   11.50
attendance_pct         75.75
marks_obtained         68.45
dtype: float64

Mode:
class            9th
section            B
subject     Computer
city       Bangalore
Name: 0, dtype: object
```

Mean, Median, and Mode of the numeric and categorical columns.

Inferences:

- The mean study_hours_per_week (15.21) is noticeably higher than the median (11.50), indicating a right-skewed distribution driven by extreme outlier values (data-entry errors of 999 hours).
- The mean marks_obtained (68.32) sits very close to the median (68.45), suggesting a roughly symmetric distribution with no major skew.
- The mode shows the most common student is in Class 9th, Section B, studies Computer, and is based in Bangalore.

Insight: The typical student is around 15 years old, studies roughly 11.5 hours a week, attends about 76% of classes, and scores around 68 marks.

Step 3: Task 2 – Descriptive Analysis – Statistical Measures of Dispersion

```
dispersion = {
    'Standard Deviation': df[num_cols].std(),
    'Variance': df[num_cols].var(),
    'Range': df[num_cols].max() - df[num_cols].min(),
    'IQR': df[num_cols].quantile(0.75) - df[num_cols].quantile(0.25)
}

print("=== Dispersion Measures ===")
for measure, values in dispersion.items():
    print(f"\n{measure}:")
    print(values)
```

```
=== Dispersion Measures ===

Standard Deviation:
age                    1.420166
study_hours_per_week    63.529121
attendance_pct          14.062910
marks_obtained          18.465786
dtype: float64

Variance:
age                      2.016871
study_hours_per_week    4035.949275
attendance_pct           197.765446
marks_obtained           340.985269
dtype: float64

Range:
age                      4.0
study_hours_per_week    1004.0
attendance_pct            49.8
marks_obtained            65.0
dtype: float64

IQR:
age                     2.000
study_hours_per_week     9.175
attendance_pct          24.700
marks_obtained          32.100
dtype: float64
```

Standard Deviation, Variance, Range, and IQR for age, study_hours_per_week, attendance_pct, and marks_obtained.

Inferences:

- study_hours_per_week has an extremely high standard deviation (63.5) and range (1004.0), far larger than its IQR (9.175) — a clear sign of a small number of severe outliers (the -5 and 999 data-entry errors).
- marks_obtained shows a moderate, well-behaved spread (std 18.47, range 65.0, IQR 32.1), consistent with natural variation in student performance.
- age has the most stable spread (std 1.42, IQR 2.0), suggesting a fairly evenly distributed student base across age groups.

Insight: The extreme range vs. IQR gap in study_hours_per_week confirms outlier handling is essential before this feature can be used reliably in modeling.

Step 4: Task 3 – Correlation Between Attributes

```
correlation_matrix = df[num_cols].corr()

print("=== Correlation Matrix ===")
print(correlation_matrix.round(4))
```

```
=== Correlation Matrix ===
                        age  study_hours_per_week  attendance_pct  \
age                  1.0000                0.0219          0.0767
study_hours_per_week  0.0219                1.0000          0.0455
attendance_pct        0.0767                0.0455          1.0000
marks_obtained       -0.0474               -0.0439         -0.0248

                      marks_obtained
age                          -0.0474
study_hours_per_week         -0.0439
attendance_pct               -0.0248
marks_obtained                1.0000
```

Pearson correlation matrix across age, study_hours_per_week, attendance_pct, and marks_obtained.

Inferences:

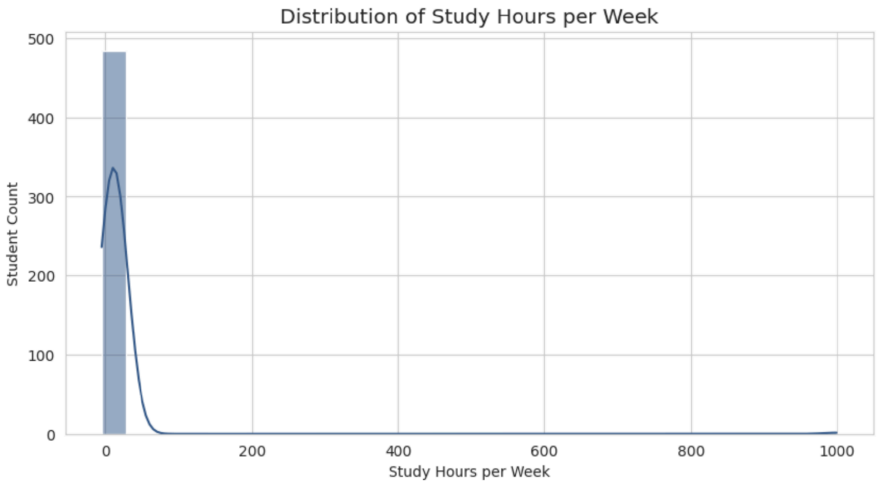
- All pairwise correlations are extremely weak (|r| < 0.08), indicating age, study_hours_per_week, attendance_pct, and marks_obtained behave largely independently of one another in this dataset.
- The near-zero correlation between study_hours_per_week and marks_obtained (-0.044) is partly explained by the outlier-inflated study_hours_per_week values distorting the linear relationship.
- Low correlation overall reduces the risk of multicollinearity, which is favourable for downstream modeling.

Insight: Once outliers in study_hours_per_week are treated, a clearer relationship with marks_obtained and attendance_pct could emerge, since consistent study habits and attendance typically support better performance.

Step 5: Task 4 – Visualization Techniques

5a. Histogram of Study Hours per Week

```
plt.figure(figsize=(10, 5))
sns.histplot(data=df, x='study_hours_per_week', bins=30, kde=True, color='#2b5c8f')
plt.title('Distribution of Study Hours per Week', fontsize=14)
plt.xlabel('Study Hours per Week')
plt.ylabel('Student Count')
plt.show()
```



Sharp right-skewed distribution — nearly all students study under 30 hours per week, with a long thin tail out to ~1,000 caused by the injected outliers.

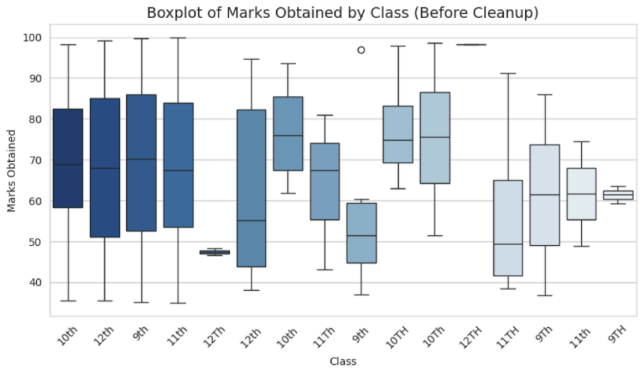
5b. Boxplot of Marks Obtained by Class (before cleanup)

```
plt.figure(figsize=(10, 5))
sns.boxplot(data=df, x='class', y='marks_obtained', palette='Blues_r')
plt.title('Boxplot of Marks Obtained by Class (Before Cleanup)', fontsize=14)
plt.xlabel('Class')
plt.ylabel('Marks Obtained')
plt.xticks(rotation=45)
plt.show()
```

```
/tmp/ipykernel_1539/1747393496.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(data=df, x='class', y='marks_obtained', palette='Blues_r')
```

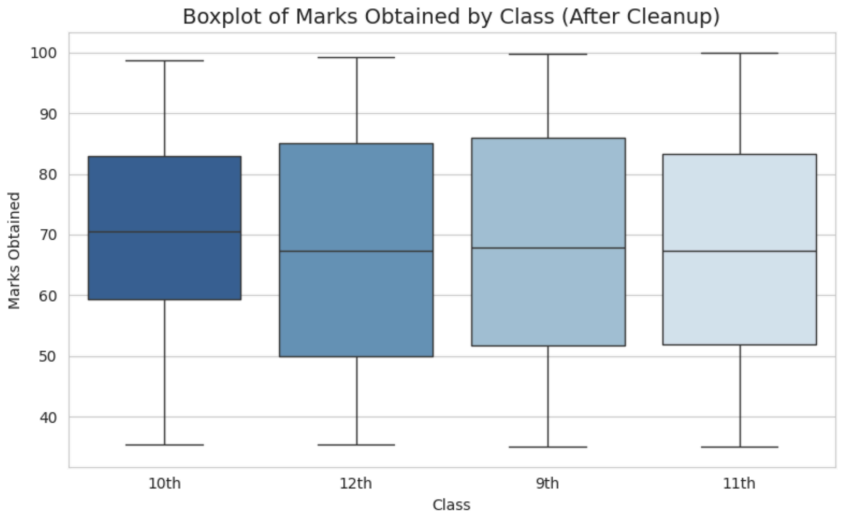


FutureWarning raised for passing palette without hue; 16 separate boxes appear because the class column still has inconsistent casing (10th / 12Th / 9TH / 10TH / etc.).

5c. Boxplot of Marks Obtained by Class (after cleanup)

```
df['class'] = df['class'].astype(str).str.strip().str.lower()

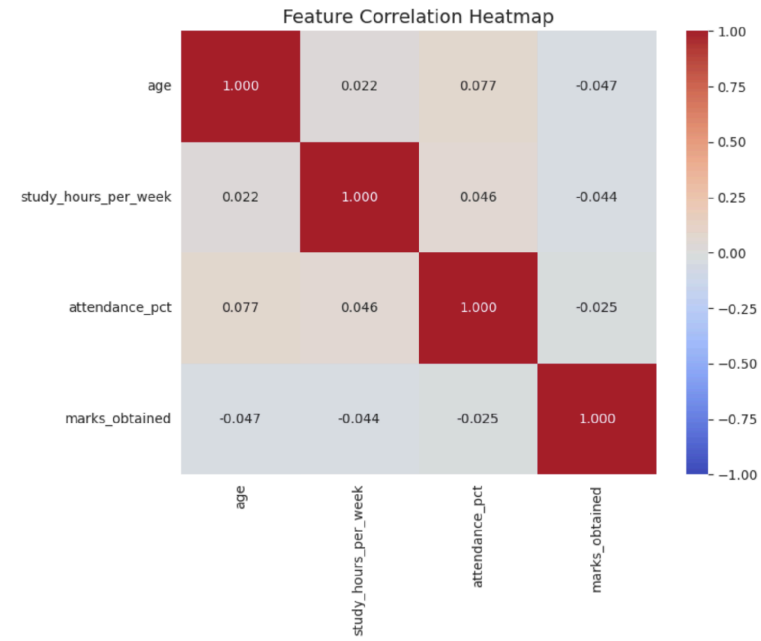
plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x='class', y='marks_obtained', hue='class', palette='Blues_r')
plt.title('Boxplot of Marks Obtained by Class (After Cleanup)', fontsize=14)
plt.xlabel('Class')
plt.ylabel('Marks Obtained')
plt.tight_layout()
plt.show()
```



After standardizing text casing, the 16 fragmented boxes collapse into the correct 4 classes — 9th, 10th, 11th, and 12th — each with a sensible, well-formed spread.

5d. Feature Correlation Heatmap

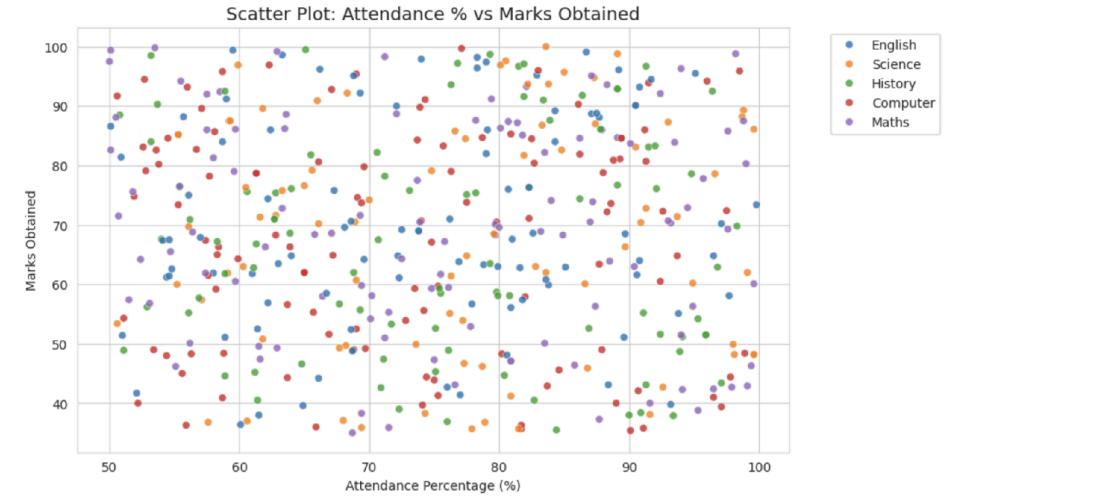
```
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', vmin=-1, vmax=1, fmt='.3f', cbar=True)
plt.title('Feature Correlation Heatmap', fontsize=14)
plt.show()
```



Heatmap confirms all pairwise correlations among age, study_hours_per_week, attendance_pct, and marks_obtained are close to zero.

5e. Scatter Plot: Attendance % vs Marks Obtained

```
plt.figure(figsize=(10, 6))
sns.scatterplot(data=df, x='attendance_pct', y='marks_obtained', hue='subject', alpha=0.8)
plt.title('Scatter Plot: Attendance % vs Marks Obtained', fontsize=14)
plt.xlabel('Attendance Percentage (%)')
plt.ylabel('Marks Obtained')
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.show()
```



Points are spread evenly across the full attendance range (50–100%) and marks range (35–100) for every subject, with no visible upward or downward trend.

Inferences:

- Histogram: study_hours_per_week is heavily right-skewed, with the vast majority of sessions under 30 hours and a small number of extreme outliers stretching the axis to 1,000 — this single-handedly explains the earlier gap between mean and median.
- Boxplot (before cleanup): inconsistent text casing in class silently created 16 categories instead of 4, which would have broken any groupby or model trained on this column.
- Boxplot (after cleanup): once standardized, marks cleanly separate by class, with all four classes showing a comparable median and no visible outliers within each group.
- Heatmap: confirms numerically what the scatter plot shows visually — no meaningful linear relationship exists between any pair of numeric features while outliers remain in the data.
- Scatter Plot: no visible trend between attendance_pct and marks_obtained for the bulk of students, regardless of subject — attendance alone does not appear to predict performance in this synthetic dataset.

Insight: Outlier treatment for study_hours_per_week and categorical text cleaning for class are the two most urgent preprocessing needs surfaced by this EDA, in addition to the missing-value and duplicate-record handling already carried out in Experiment 2.

Conclusion

The exploratory data analysis (EDA) on the Student Result dataset provided valuable insights into the characteristics and relationships of student demographic and performance features. Measures of central tendency and dispersion revealed that study_hours_per_week is heavily distorted by a small number of extreme outliers, while marks_obtained, attendance_pct, and age show reasonably well-behaved distributions.

Correlation analysis found only weak relationships ($|r| < 0.08$) among all numeric features, largely because the unresolved outliers in study_hours_per_week mask any genuine association with marks_obtained or attendance_pct.

These findings enhance the dataset's quality for machine learning by identifying critical features (study_hours_per_week, attendance_pct, class) and highlighting the exact preprocessing steps — outlier treatment and categorical text standardization — needed before the data can be reliably used for predictive modeling.

DMBI Laboratory - Experiment 3

Experiment 3: Exploratory Data Analysis and Visualization

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Division: D15A

Course / Subject: Data Mining and Business Intelligence (DMBI)

Academic Year: 2026 - 2027

Official GitHub Experiment & Submission Links:

1. Individual GitHub Link for Experiment 3:

https://github.com/wagha-builds/dmbi_exps_59/tree/main/Experiment_3

2. Master GitHub Repository Link:

https://github.com/wagha-builds/dmbi_exps_59

Note: This document has been submitted as soft copy and printed as hard copy per lab guidelines. All supporting datasets, code files, and analytical outputs are maintained at the GitHub links above.